Tuning report: tropical / SCC

2026-10-22T19:17:34

1 Decision

Approving freezes `fishr.lambda` = 1000 into every cell of the grid (frozen hash `d8981f9390da`). The experiment YAML has `fishr.lambda` = 1000 for this campaign. `frozen.yaml` is currently unapproved with {'fishr.lambda': 1000.0}.

Selection rule. The candidate with the lowest mean `id_val_loss` wins, averaged over 6 cell(s) and 2 tuning seed(s) each (global seed 1000, disjoint from the grid's 0), across 4 candidates. A candidate with a missing or diverged run is not eligible.

Verdict: REVIEW

- **review**: the winner is not distinguishable from fishr.lambda=100: the paired gap's 95% CI is [-6.06e-05, 6.91e-05]
- **review**: fishr.lambda=10: fishr.lambda * fishr_penalty_final is 0.0014 of id_val_loss (median over runs); the term barely enters the loss
- **review**: 12 of 12 winner runs reached their best id_val_loss at the last logged epoch: the search budget may be too short to rank candidates
- **review**: 12 of 12 winner runs picked their last epoch (best_epoch = 19): still improving when the search budget ran out, so candidates were ranked before converging
- **info**: the search ran with optim.epochs=20; the grid runs with optim.epochs=100
- **info**: devices: cuda:0

block: do not approve; fix the search and re-run it. **review**: read the sections below before approving. **info**: context only.

2 Search setup

- `fishr.lambda` ∈ {10, 100, 1000, 1e+04}
- Cells trained (`tune.only` {'objective': 'fishr'}):
  - `model_type=vanilla head=seq2seq objective=fishr`
  - `model_type=vanilla head=pointer objective=fishr`
  - `model_type=adaptive head=seq2seq objective=fishr`
  - `model_type=adaptive head=pointer objective=fishr`
  - `model_type=tropical head=seq2seq objective=fishr`
  - `model_type=tropical head=pointer objective=fishr`
- Search-only overrides (never frozen): {'optim': {'epochs': 20}}. The chosen value was ranked under this budget, not the grid's.

3 Candidates

Table 1 ranks the candidates on `id_val_loss` (lower is better); Figure 1 plots them. Candidates share seeds, so Table 2 compares each with the winner run by run (same cell, same seed).

Table 1: Every candidate: rank, runs stored/expected, diverged, mean with 95% bootstrap CI, SD and range over all its runs.

#	fishr.lambda	runs	div.	mean	95% CI	SD	range	
3	10	12/12	0	0.003319	[0.00307, 0.00362]	5.17×10^{-4}	[0.00279, 0.00438]	
2	100	12/12	0	0.003284	[0.00304, 0.00354]	4.80×10^{-4}	[0.00263, 0.00421]	
1	1000	12/12	0	0.003281	[0.00302, 0.00357]	5.19×10^{-4}	[0.00256, 0.00436]	**selected**
4	1e+04	12/12	0	0.00341	[0.00317, 0.00369]	4.78×10^{-4}	[0.00295, 0.00442]	

Table 2: Each candidate minus the winner on the runs they share, signed so a positive gap is worse than the winner. Worse/better/equal count the runs; p is a two-sided sign test.

candidate	n	gap	95% CI	worse	better	equal	p
`fishr.lambda=10`	12	3.875×10^{-5}	$[-1.19 \times 10^{-5}, 9.90 \times 10^{-5}]$	7	5	0	0.7744
`fishr.lambda=100`	12	3.363×10^{-6}	$[-6.06 \times 10^{-5}, 6.91 \times 10^{-5}]$	7	5	0	0.7744
`fishr.lambda=10000`	12	1.298×10^{-4}	$[6.44 \times 10^{-5}, 2.07 \times 10^{-4}]$	10	2	0	0.0386

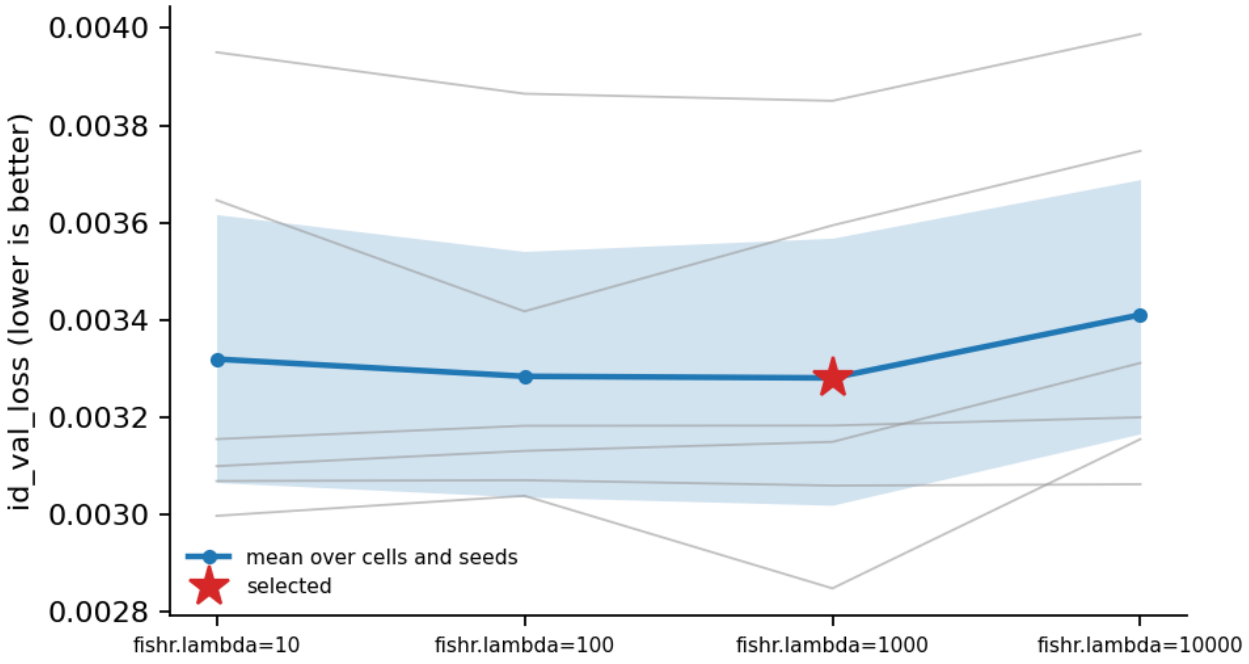


Figure 1: `id_val_loss` per candidate: the mean over every run with a 95% bootstrap CI (blue), each cell's seed-mean (grey), the selected candidate (red star).

4 Per cell

One value is frozen for every cell, chosen by the mean over cells. Figure 2 shows whether the winner is best in each cell or is carried by a few of them.

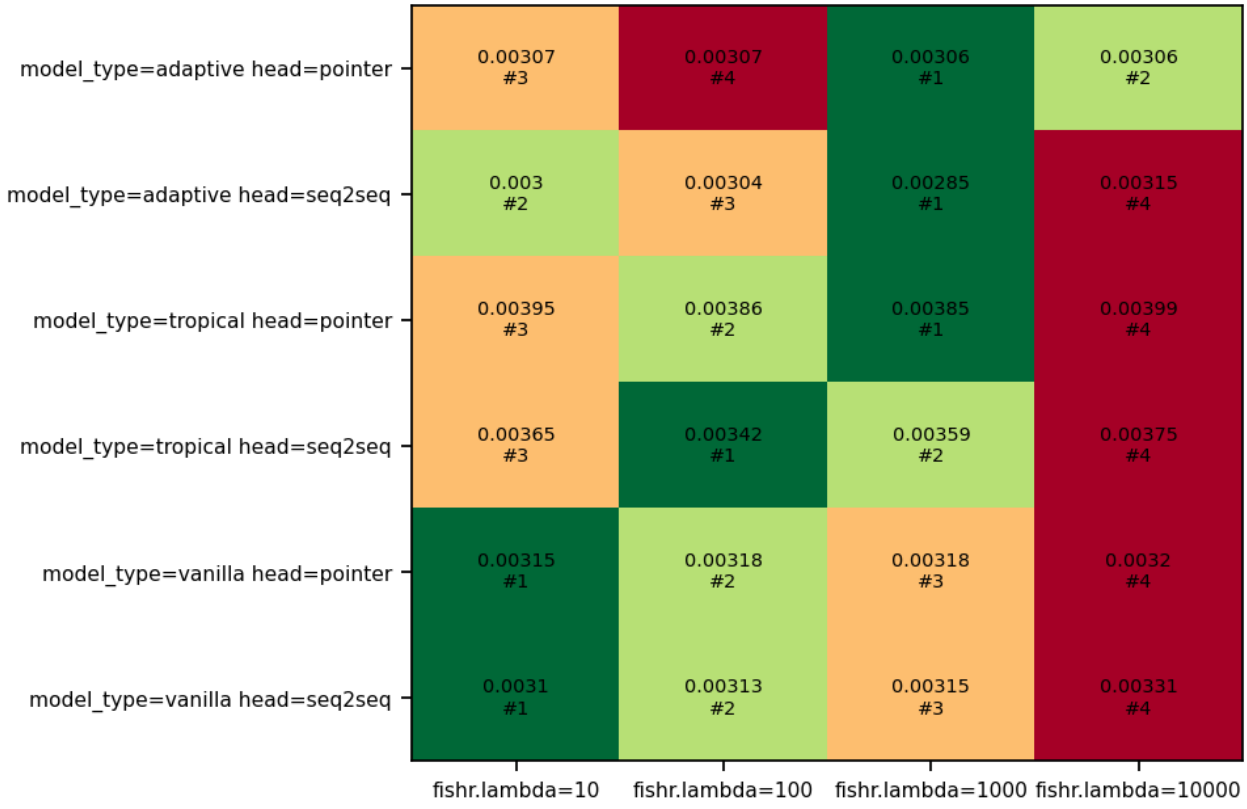


Figure 2: Each cell's seed-mean `id_val_loss` per candidate, with its rank within the cell (#1 is best; green is better).

Table 3: The best candidate in each cell.

cell	best
`model_type=adaptive head=pointer`	fishr.lambda=1000
`model_type=adaptive head=seq2seq`	fishr.lambda=1000
`model_type=tropical head=pointer`	fishr.lambda=1000
`model_type=tropical head=seq2seq`	fishr.lambda=100
`model_type=vanilla head=pointer`	fishr.lambda=10
`model_type=vanilla head=seq2seq`	fishr.lambda=10

5 Other training-distribution metrics

Not used for selection; shown so a winner that trades them away is visible.

Table 4: Mean of every other in-distribution metric per candidate.

candidate	`id_f1`	`id_exact`	`fishr_penalty_final`	`best_epoch`
fishr.lambda=10	0.996	0.979	1.04×10^{-6}	19

Table 4: Mean of every other in-distribution metric per candidate.

candidate	id_f1	id_exact	fishr_penalty_final	best_epoch
fishr.lambda=100	0.996	0.979	4.75×10^{-7}	19
fishr.lambda=1000	0.996	0.979	1.29×10^{-7}	19
fishr.lambda=10000	0.996	0.979	4.12×10^{-8}	19

6 Training curves

The mean over every cell and seed of each candidate; the selected candidate in bold red.

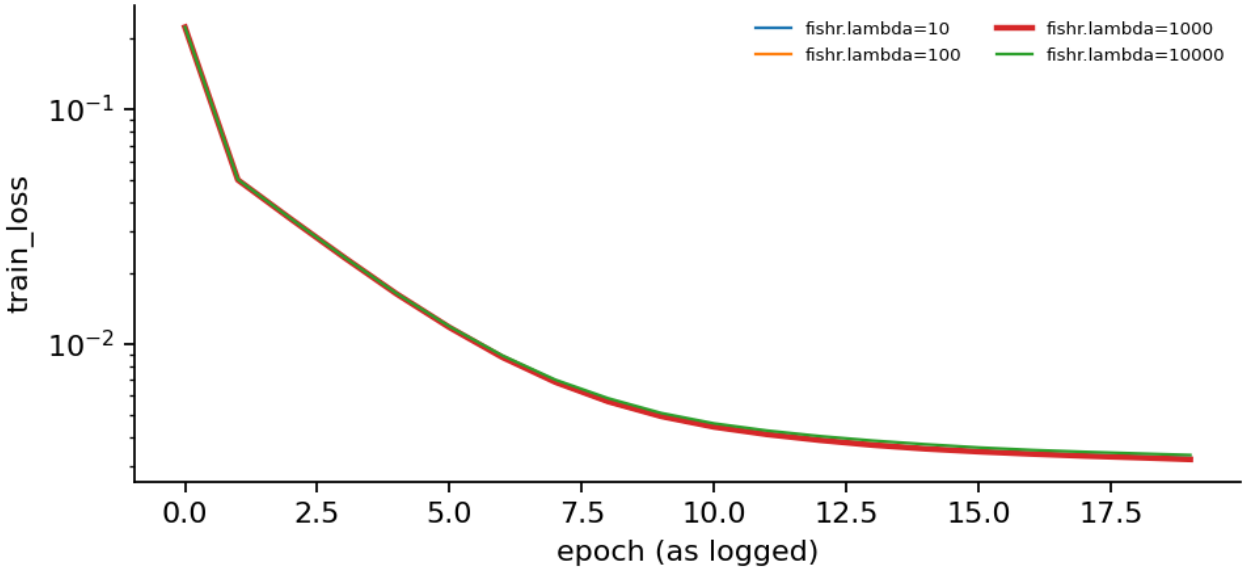


Figure 3: `train_loss` per candidate.

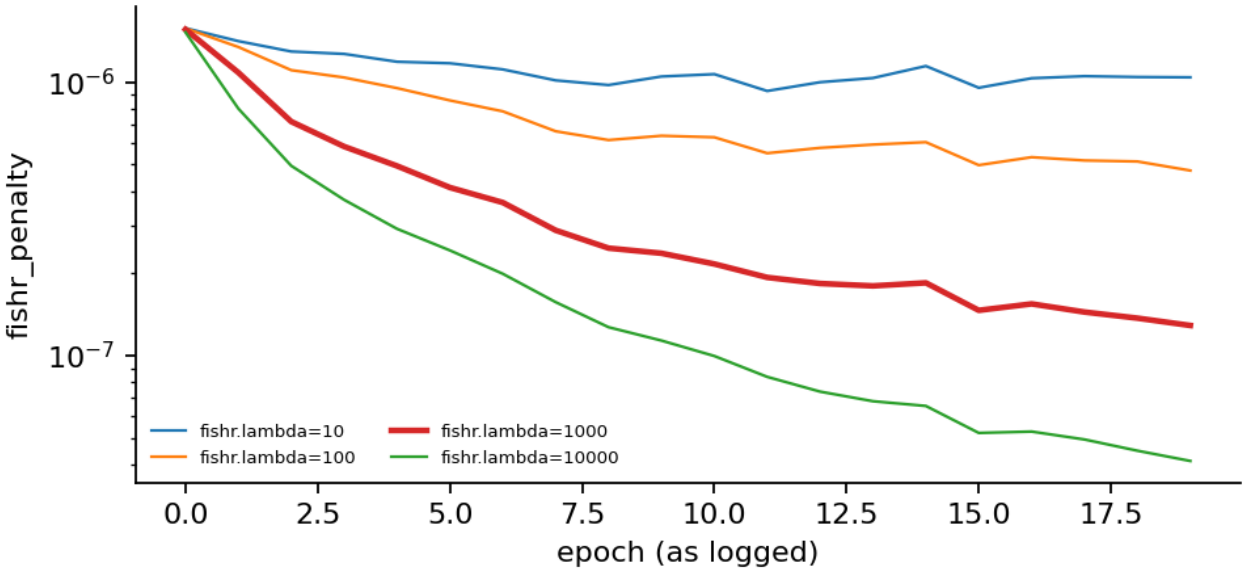


Figure 4: `fishr_penalty` per candidate.

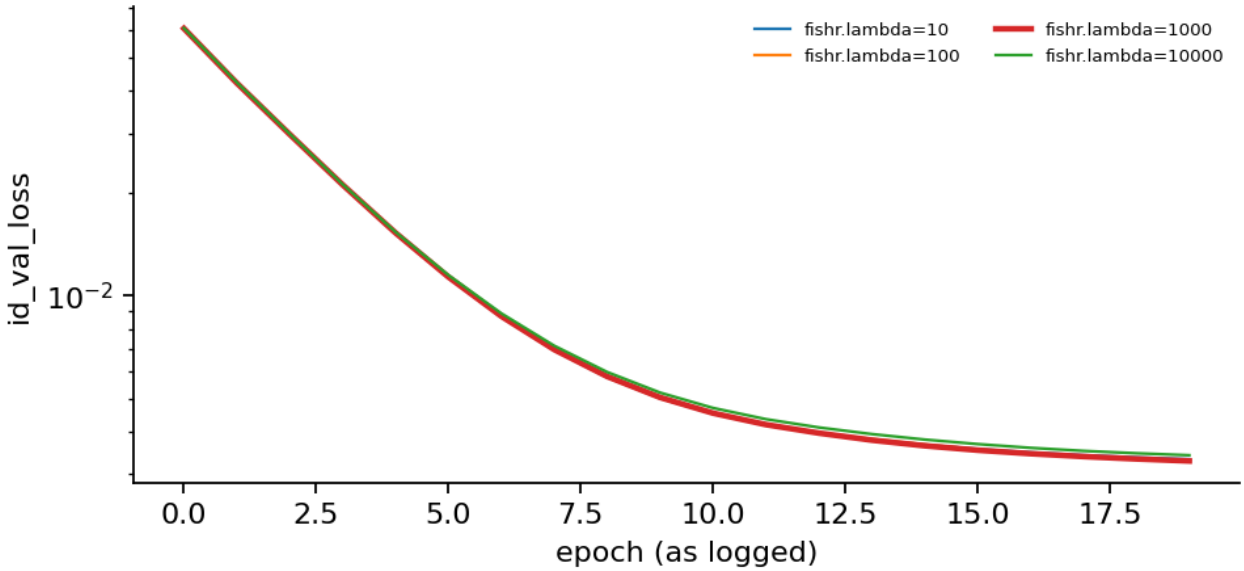


Figure 5: `id_val_loss` per candidate.

7 Provenance

Table 5: Each candidate's runs: base hash, runs, total wall time, commits (+ marks a dirty tree), torch.

candidate	hash	runs	wall (s)	commit	torch
fishr.lambda=10	`b7150148661b`	12	3.51e+04	`e5ccce8034`	2.8.0+cu128
fishr.lambda=100	`7984c1850aea`	12	2.41e+04	`e5ccce8034`	2.8.0+cu128
fishr.lambda=1000	`52ab10df6885`	12	2.04e+04	`e5ccce8034`	2.8.0+cu128
fishr.lambda=10000	`5a197ec9fc42`	12	3.78e+04	`e5ccce8034`	2.8.0+cu128

Store: `campaigns/tropical/SCC/tune/results`.
Rebuild: `expkit tune-report --config configs/tropical.yaml --profile SCC`
Approve: `expkit approve --config configs/tropical.yaml --profile SCC`

The base the grid would run with.

```
task:
  name: SCCDataset
  value_range: 0.001
  target_range: null
  weight_range: null
  adversarial_range: null
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  ood_value_range: 0.1
  ood_noise_range: null
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 8
  eval_batch_size: 8
fishr:
  lambda: 1000.0
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

A Sealed: OOD and test metrics

Read this only after deciding. These were recorded by the tuning runs but never consulted by the selection. Choosing a value because it looks better here would tune on the test distribution.

Table 6: Mean of every OOD or test metric per candidate.

candidate	ood_len_f1	ood_len_exact	ood_val_f1	ood_val_exact	ood_noise_f1	ood_noise_exact	ood_len_f1_L12	ood_len_exact_L12	ood_len_f1_L16	ood_len_exact_L16
fishr.lambda=10	0.649	0.165	0.965	0.87	0.722	0.321	0.79	0.31	0.576	0.0204
fishr.lambda=100	0.644	0.135	0.965	0.87	0.706	0.312	0.784	0.249	0.571	0.0218
fishr.lambda=1000	0.652	0.131	0.965	0.868	0.707	0.321	0.782	0.24	0.583	0.0232
fishr.lambda=10000	0.679	0.227	0.965	0.869	0.716	0.336	0.782	0.371	0.622	0.0834